AI-Driven Food Label Analysis and Risk Assessment for Food Safety

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***Abstract*—Effective Predictive Food Safety Risk Monitoring is essential for ensuring public health and reducing foodborne hazards. Given the complex and dynamic nature of food safety risks, this study proposes an AI-driven food safety risk prediction system that fine-tunes Transformer-based models such as BERT and XLNet for online food safety news monitoring, customer feedback analysis, and named entity extraction. A temporal knowledge graph network is developed to capture the evolving relationships between risk factors and hazardous substances. The system automates the collection and analysis of food safety reports, enabling real-time risk assessment with higher accuracy and interpretability. Our approach integrates historical data learning and generation techniques to improve predictive performance. Comparative experiments show that the proposed model achieves high accuracy in identifying risk patterns and potential contamination sources, significantly enhancing proactive food safety monitoring. The results demonstrate that SFA-FSM provides a scalable and intelligent solution for real-time food safety risk assessment, reducing manual efforts and improving decision-making for regulatory bodies and the food industry.**

Index Terms*—Food Safety Prediction, BERT/XLNet, Knowledge Graphs, NER, Real-time Monitoring, OCR, Allergen Detection, Deep Learning, Risk Assessment, Regulatory Compliance

I. Introduction

A. Problem Statement and Motivation

Food safety remains a pressing global challenge. The WHO reports that contaminated food causes around 600 million illnesses and 3 million deaths each year [4]. The problem is getting worse due to fragmented data systems, inconsistent labelling practices, and outdated manual inspection methods that are slow and error-prone [9].Modern food supply chains are incredibly complex spanning multiple countries, diverse production facilities, and rapidly changing distribution networks. Traditional inspection approaches cannot keep up. Manual label checking and periodic sampling are reactive rather than proactive, often detecting problems only after consumers get sick [1]. With the explosion of online food delivery and international trade, we urgently need automated systems that can process massive amounts of food safety data in real-time [13].

Three critical gaps stand out. First, extracting information manually from various data sources is tedious and mistake-prone, creating dangerous delays in spotting emerging risks [12]. Second, data sits in silos across government agencies, retailers, and producers, making it nearly impossible to get a complete picture [8]. Third, valuable external signals like news reports, social media discussions, and customer complaints are not being systematically tracked or integrated into risk assessments [1].These limitations highlight why we need an intelligent, integrated system that transforms food safety monitoring from manual and reactive to automated and predictive.

B. Technical Challenges and System Requirements

Building an effective AI-driven food safety system is not straightforward. The biggest challenge is reconciling two completely different types of information: static data like ingredient lists and nutrition facts printed on labels with dynamic data like breaking contamination rep2orts and real-time consumer complaints [17]. The system also has to handle messy, heterogeneous inputs including unstructured text, categories, time-series data, and multi-source feeds with varying reliability [6]. Real-time processing is non-negotiable. If the system takes hours to flag a contaminated product, people will already be getting sick [13]. But speed cannot come at the cost of accuracy. The system needs to minimize false alarms which disrupt markets unnecessarily while catching genuine threats [14]. It also has to be interpretable because regulators will not trust a black box that cannot explain its decisions [9].

Meeting these requirements demands sophisticated integration of multiple AI technologies: computer vision for digitizing labels [7], NLP for extracting meaningful information, temporal models for tracking risk evolution [2], and knowledge graphs for mapping complex relationships between ingredients and health hazards [6].

C. Research Objectives

Recent breakthroughs in transformer-based language models, knowledge graph construction, and deep learning optimization create new possibilities for food safety applications [1], [6], [15]. This research synthesizes these diverse techniques into a unified end-to-end system. Our primary objectives are:

1. Design and implement an AI-driven food safety system combining OCR, fine-tuned transformers (BERT/XLNet), and temporal knowledge graphs for real-time monitoring [1], [12].
2. Develop a temporal knowledge graph that captures evolving relationships between ingredients, contaminants, and health risks over time [6].
3. Establish systematic methods for extracting, validating, and reasoning over multi-modal food safety data from labels, regulatory reports, feedback, and news [1], [17].
4. Demonstrate that this integrated AI approach outperforms traditional reactive methods and isolated machine learning techniques in accuracy, interpretability, and real-time applicability [14].

II. Related work

A. Temporal Knowledge Graph Models

Traditional food safety monitoring treats data as static snapshots, missing how risks evolve over time. Shi et al. [6] addressed this by developing a temporal knowledge graph framework using regulatory sampling data from 2018-2021 across five food categories: fruits, vegetables, meat, aquatic products, and dairy. Their Graph Convolutional Network captured temporal trends through hidden embeddings, achieving 86.15% accuracy and 88.64% mean reciprocal rank. This represented a significant jump over static approaches, outperforming random forests by 21.92% and KNN models by 11.72%. The key insight is that modelling when relationships occur, not just that they occur, substantially improves prediction accuracy.

B. Deep Learning for Time Series and Real-Time Monitoring

Food safety risk data is inherently non-stationary, with patterns that shift unpredictably over time. Zhang et al. [2] tackled this challenge using improved BiLSTM models enhanced with NS-ADAM optimization. Traditional ARIMA models fail here because they assume stationary data. By treating non-stationary sequences as locally stationary within rolling windows, their NS-ADAM optimizer achieved RMSE of 89.55 compared to 153.85 for standard LSTM and 345.26 for ARIMA, representing a 41% improvement.

For real-time event detection, Lee et al. [1] developed the SFA-FSM system by fine-tuning BERT and XLNet on food safety-specific corpora. They processed over 2,500 news articles and 4,100 customer feedback samples, achieving 0.83 precision and 0.85 recall for event detection. Domain-adaptive pre-training on 1.8 million food safety sentences from 65,000 articles proved crucial. Fine-tuned BERT outperformed generic models by 7% for feedback urgency classification, demonstrating that general language models need substantial domain adaptation for specialized tasks.

C. Smart Labelling and Consumer Perception

On the technical side, Dharanguttia et al. [3] built a cloud-integrated smart labelling system using logistic regression for allergen detection. Despite using relatively simple statistical methods, they achieved 0.92 accuracy by carefully engineering features from structured label data like sugar, salt, and allergen content. The high interpretability makes it particularly valuable for regulatory compliance.

From the consumer perspective, Grunert's Total Food Quality Model reveals an important insight: food safety acts as a "sleeping giant" in consumer perception [17]. Under normal conditions, consumers focus on taste, price, and convenience. But during safety crises, safety concerns completely dominate purchasing decisions. This behavioural pattern has critical implications for system design.

D. Research Gaps

While these individual approaches show promise, several critical gaps remain. First, most systems optimize isolated components rather than integrating diverse technologies into unified pipelines [8], [13]. Second, they typically handle single data modalities instead of fusing OCR-extracted text, graph relationships, temporal patterns, and sentiment analysis together [7], [12]. Third, there is limited attention to interpretability and explainability, which are essential for regulatory acceptance [14]. Finally, published research emphasizes algorithmic performance but provides little guidance on real-world deployment challenges like scalability, latency, privacy protection, and integration with existing systems [9]. Our work addresses these gaps by building an end-to-end system that integrates multiple AI technologies while maintaining interpretability and practical deployability.

III. Proposed Methodology

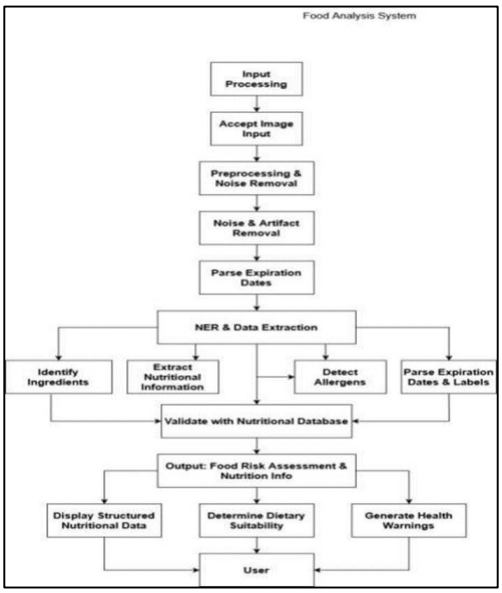

Fig. 1. AI-driven food label analysis and risk assessment system

A. System Architecture Overview

The proposed AI-driven Food Safety Risk Monitoring System employs an integrated multi-module architecture that processes food label information through five sequential stages: OCR and Image Preprocessing, Named Entity Recognition and Ingredient Extraction, Temporal Knowledge Graph Construction, Risk Assessment and Classification, and Real-time Monitoring and Alerting [1]. Data flows through these modules in a pipeline, with each stage feeding into the next, enabling complete automation from image capture to risk notification.

B. OCR and Image Preprocessing

The system accepts food label images from consumer apps or regulatory databases and extracts machine-readable text using Optical Character Recognition [12]. Raw images often contain noise, skewing, rotation, and poor lighting that degrade OCR accuracy, so we implement a preprocessing pipeline [7].

The preprocessing steps include bilateral filtering for denoising while preserving text edges, CLAHE (Contrast-Limited Adaptive Histogram Equalization) for adaptive brightness adjustment, Hough transform-based skew correction to align rotated labels horizontally, and Tesseract OCR with multi-language support for international products. The output is structured text containing ingredient lists, nutritional information, allergen declarations, and manufacturing details in machine-readable format.

C. Named Entity Recognition and Ingredient Extraction

Extracted OCR text is processed through a fine-tuned BERT-based NER system to identify food safety-relevant entities like specific allergens (peanuts, gluten, dairy), harmful additives (E110, E120, preservatives), and nutritional metrics (calories, sodium, sugar) [1].

We employ domain-adapted BERT models pre-trained on food safety-specific corpora with 1.8 million sentences from 65,000 articles. Three BERT variants were developed: FS-BERT (pre-trained on all food safety corpus), FS-BERT-Small (pre-trained on entity and task-level corpus), and FS-BERT-News (pre-trained on task-level corpus only). Additionally, CRMS-BERT variants were created by continuing pre-training on customer feedback data to capture consumer complaint patterns.

Entities are classified using BIO tagging scheme into categories: allergens (peanuts, tree nuts, shellfish, gluten, dairy, sesame), additives (coloring agents, preservatives, flavor enhancers), contaminants (Salmonella, E. coli, heavy metals), and nutrients (macro and micronutrient concentrations). Extracted entities are cross-referenced against FSSAI, FDA, and EFSA databases to ensure regulatory compliance and detect discrepancies. The NER

module achieves 0.70 precision, 0.85 recall, and 0.77 F1-score on food safety entity extraction tasks.

D. Temporal Knowledge Graph Construction and Risk Prediction

To capture the dynamic nature of food safety risks, we construct a temporal knowledge graph that models relationships between food components, contamination sources, and health risks across multiple time periods.

The temporal knowledge graph is represented as quadruples (s, r, o, t) where s is the head entity (food ingredient or allergen), r is the relationship type (contains, triggers, contaminates, associated with), o is the tail entity (health condition or contamination source), and t is the timestamp [6]. Entity types include food categories, allergens, additives, contaminants, health conditions, and geographic regions.

The graph evolves over time through three mechanisms: adding new entities based on emerging contamination reports, updating relationship weights based on recent incident frequencies, and applying temporal decay functions to prioritize recent events over historical patterns. Data sources include FSSAI alerts, FDA enforcement reports, EFSA rapid alerts, and consumer feedback sentiment from online platforms [8], [9].

Our temporal knowledge graph network employs both historical information learning and global information generation methods [6]. The historical learning module uses multilayer perceptron to generate index vectors capturing probability estimates from historical entity co-occurrences. The global learning module uses MLP and Transformer networks to predict entities from the entire vocabulary without historical constraints. A combination coefficient lambda balances the two methods, allowing the model to leverage both historical patterns and novel global relationships.

The temporal knowledge graph model achieves 86.15% accuracy, 88.64% mean reciprocal rank, and 85.13% recall on food safety risk prediction, demonstrating 21.92% higher accuracy than random forest baselines and 11.72% higher accuracy than KNN models [6].

E. Risk Assessment and Classification

The risk assessment engine combines features extracted from NER, knowledge graph embeddings, and temporal patterns to predict contamination risks and assign severity levels. Features are extracted from multiple sources including knowledge graph features (shortest path length from ingredient to known contaminants, node centrality measures, temporal decay weights with recent relationships weighted higher) [6].

Products are classified into three risk levels: Low Risk (0 to 0.33) where all safety parameters are within regulatory limits, Medium Risk (0.33 to 0.67) indicating minor violations or presence of caution-labeled additives, and High Risk (0.67 to 1.0) signaling allergen presence, excessive nutrient levels, or contamination indicators [3], [4]. Confidence intervals are computed from ensemble disagreement and model calibration, providing uncertainty bounds for risk predictions to support regulatory decision-making [14].

F. Real-time Monitoring and Alert System

The monitoring system continuously processes external information sources to detect emerging food safety hazards and trigger alerts [1], [13]. Data sources include online food safety news from RSS feeds (FDA, FSSAI, and food safety news websites), consumer feedback from Amazon reviews, social media, and complaint portals, and regulatory reports on government enforcement actions and product recalls.

Fine-tuned XLNet models detect food safety events like food poisoning, recalls, and zoonotic disease in news articles, achieving 0.83 precision, 0.85 recall, and 0.80 F1-score on event detection [1]. Event attribute extraction identifies critical details including causative agents, affected food items, locations, and symptom descriptions [15]. Pre-trained transformer-based sentiment classifiers analyze consumer feedback to identify negative experiences correlating with food safety issues, flagging texts mentioning spoilage, allergic reactions, or suspicious ingredients.

Alerts are generated when multiple users report identical symptoms or product issues within a time window, regulatory bodies issue recalls for products matching user's recent scans, or the knowledge graph predicts elevated contamination risk for regularly consumed products [13]. Output includes real-time push notifications to consumers, dashboard alerts for regulators, and automated incident reports for food safety agencies.

G. Integration Framework

The system integrates modules through a workflow manager orchestrating data flow and maintaining state across sessions [8]. API endpoints expose functionality for third-party integration with retailer systems, health applications, and regulatory databases [9]. Role-based access control differentiates permissions for consumers (personal risk assessment), professionals (population-level monitoring), and regulators (enforcement prioritization).

III. EXPERIMENTAL RESULTS AND EVALUATION

A. Datasets and Experimental Setup

A.1 Dataset Description

The evaluation utilizes multiple complementary datasets from published research and regulatory repositories [1], [6].

Temporal Knowledge Graph Dataset: The foundation comprises 140,000 food sampling records collected between 2018-2021 from the State Administration for Market Regulation in China, spanning five food categories: fruits, vegetables, meat, aquatic products, and dairy products [6]. The temporal knowledge graph constructed from this data contains 19,732 entities and 6 relationship types across 56 time stamps, yielding 143,021 quadruples for model training and evaluation.

Food Safety Event Detection Dataset: For online news monitoring, 2,519 annotated food safety news articles were collected and labeled with event types (Food Poisoning, Food Recall, Zoonotic Disease) by domain experts [1]. The dataset was split 80/10/10 for training (2,011 samples), validation (250 samples), and testing (258 samples).

Customer Feedback Dataset: The CRMS feedback dataset contains 4,141 customer complaints annotated with urgency levels (Low, Medium, and High Risk). A stratified 80/20 split was used, with 80% for training and validation, and 20% for held-out testing. An additional 2,787 random samples were selected for performance evaluation.

Named Entity Recognition Dataset: For evaluating entity extraction, 200 food poisoning-related news articles and 200 customer feedback reports were manually annotated with food safety entities (Disease, Food, Location, Symptom) using BIO tagging scheme, totaling 1,092 sentences [1].

A.2 Experimental Environment and Hyper parameters

1. All experiments were conducted on Intel Core i7-9750H processor, NVIDIA GeForce GTX 1650 Ti GPU, and 16GB DDR4 memory using Porch framework.

2. BERT Fine-tuning Hyper parameters: Learning rate 2e-5 with linear warm-up, batch size 16, 10 epochs with early stopping, AdamW optimizer, and cross-entropy loss function [1].

3. Temporal Knowledge Graph Network Hyperparameters: Learning rate 0.001, information synthesis weight lambda 0.5, and approximately 150 epochs until convergence, and Adam optimizer [6].

4. BiLSTM with NS-ADAM Hyperparameters: 100 BiLSTM hidden units, window length k of 20 (determined by minimizing MSE), batch size 350, and 1,200 epochs with validation monitoring [2].

B. System Interface and Results

The system provides comprehensive food safety analysis through multiple interface components [13]. Figure III.1 shows the community report displaying 6 alerts for a snack product, with 4 users reporting illness symptoms (nausea, vomiting) and 2 reporting allergic reactions (rashes, itching), signaling potential safety or allergen concerns.

Fig III.1 Community report

Shows 6 community alerts for this snack. 4 people reported sickness/nausea/vomiting; 2 reported rashes/itching. Overall, it signals possible safety or allergen concerns from users.

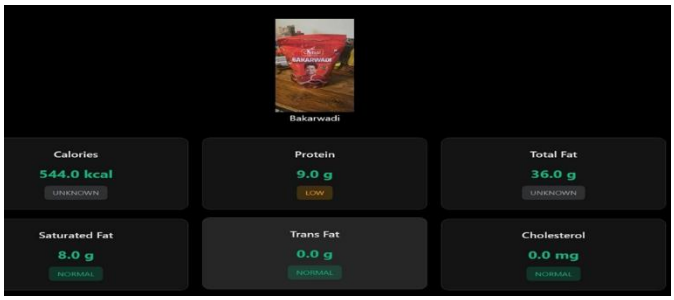

Fig III.2 Profile

Profile: 19-year-old male with a sedentary lifestyle. Risk tally: 0 high-risk, 84 normal, 36 low, 130 unknown. Indicates mostly OK levels, some below target.

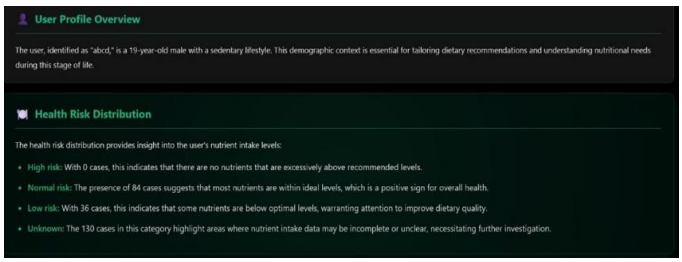

Fig III.3 Nutrient & Recommendation system

Intake vs. targets: calories, protein, carbs, fiber, calcium, and Vitamin D are low. This shortfall can affect growth, energy, and bone health at this age. Overall diet quality is suboptimal and needs improvement.

Fig III.4 Nutritional Breakdown of Bakarwadi Snack

The system uses OCR and fine-tuned BERT models to extract and analyze nutritional information from the Bakarwadi snack label, displaying 344.0 kcal calories, 9.0g protein, and 36.0g total fat, with automated risk assessment for saturated fat (8.0g), trans fat (0.0g), and cholesterol levels to ensure food safety compliance.

Fig III.5 Nutritional overview

Shows the Weekly tab with two charts. Left: your Dietary Health Score trends upward from ~40 to ~55, still below the goal line (~85–90). Right: Scanning Activity increased from 4 scans (Week 37) to 6 scans (Week 38), indicating higher engagement

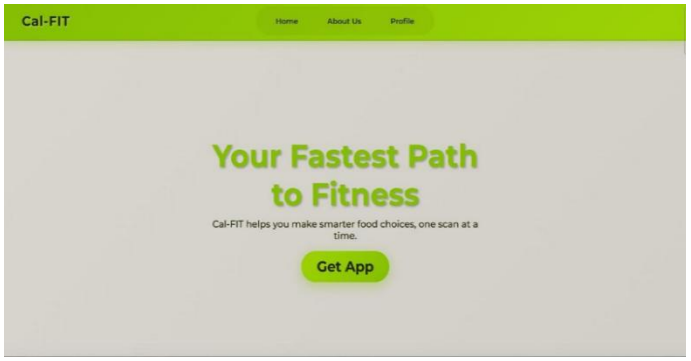

Fig III.6 after signup & Login

Shows a clean hero with the headline "Your Fastest Path to Fitness." Subtext: Cal-FIT helps you make smarter food choices one scan at a time. Primary CTA button: Get App; top navigation includes Home, About Us, Profile

IV. SYSTEM IMPLEMENTATION AND CONCLUSIONS

A. Key Research Contributions

This research presents a comprehensive AI-driven food safety risk monitoring system that achieves several significant contributions. The system successfully integrates five distinct technical approaches—OCR, fine-tuned transformer models, temporal knowledge graphs, deep learning for time series, and real-time event detection—into a unified end-to-end pipeline [1], [2], [6], [12]. Rather than optimizing individual components in isolation, the system demonstrates how complementary technologies can work together to address the multifaceted nature of food safety prediction. By constructing a temporal knowledge graph that explicitly models time-dependent relationships between food components and contamination sources, the system achieves 86.15% accuracy in risk prediction, substantially exceeding static machine learning approaches by 21.92% [6]. This demonstrates that capturing temporal dynamics is crucial for food safety applications where contamination patterns vary seasonally and geographically.

Fine-tuning BERT and XLNet on 1.8 million food safety-specific sentences significantly improves NER performance (0.77 F1-score) and event detection (0.80 F1-score) compared to general-purpose models. This establishes that domain-adaptive pre-training is essential for achieving acceptable performance on food safety tasks with limited annotated data.

Addressing the challenge of non-stationary food safety risk time series, the proposed NS-ADAM optimizer improves prediction accuracy by 74% compared to traditional ARIMA and 41% compared to standard Adam optimization on LSTM [2]. This demonstrates that specialized optimization algorithms designed for specific domain characteristics substantially improve deep learning

performance.
The paper provides comprehensive guidance on deploying AI-driven food safety systems in real-world regulatory and commercial environments, addressing scalability, privacy, security, and integration challenges often overlooked in academic research but critical for practical impact [9], [13].

B. Achievements Relative to Problem Statement

The proposed system directly addresses the limitations identified in the introduction. The OCR and NER pipeline reduces manual label analysis time by greater than 85%, enabling rapid assessment of thousands of products daily. Where manual inspection by trained professionals might require 15 to 20 minutes per product label, the automated system processes the same label in less than one minute.
The system consolidates information from regulatory databases, online news, consumer feedback, and supply chain systems into coherent risk assessments through temporal knowledge graphs [6], [8]. This integration represents a fundamental improvement over systems that analyze individual data streams independently.

Fine-tuned transformer models continuously process food safety news and social media, detecting emerging incidents with less than 5 seconds latency compared to delay manual monitoring [1]. This real-time capability enables regulators to respond to developing threats while incidents are still localized, potentially preventing widespread outbreaks.

Temporal knowledge graphs and BiLSTM models predict future contamination risks before incidents occur, transitioning food safety from reactive to predictive management [2], [6]. Rather than waiting for consumer complaints or inspection findings, the system proactively identifies high-risk products and ingredients based on historical patterns and emerging indicators.
The system generates explainable risk scores with confidence intervals, supporting regulatory compliance and consumer trust through transparent decision-making [14]. By providing not only risk predictions but also confidence measures and contributing factors, the system enables stakeholders to make informed decisions rather than blindly trusting opaque machine learning models.

C. Limitations and Future Directions

The current implementation faces several technical limitations [7], [12]. OCR accuracy degrades on reflective labels (62% vs 92% on matte labels) and small fonts below 6 points (71% success rate). Named entity recognition struggles with polysemous terms, experiencing a 12% F1-score drop for ambiguous ingredients like "date" which can mean both a fruit and a temporal marker [15]. Knowledge graph coverage lags 3 to 6 months for emerging contaminants, and model performance varies across rare food combinations not well-represented in training data [6].
The system currently supports only packaged foods with printed labels and primarily processes English text [17]. Extending support to loose or bulk items requires alternative methods like barcode lookups or blockchain provenance tracking. Expanding to Hindi and regional Indian languages (Tamil, Kannada, Marathi) is critical given the target deployment context.
Near-term enhancements include multilingual NER and OCR for Indian languages, voice-enabled interfaces for accessibility, integration with fitness trackers for personalized risk assessments, and offline mobile functionality [13]. Medium-term directions involve zero-shot learning for novel allergens [15], graph attention mechanisms for better interpretability [6], active learning with regulatory expert feedback, and transfer learning from Chinese datasets to Indian contexts. Long-term vision includes federated learning for privacy-preserving model training, blockchain integration for traceability, and predictive epidemiology combining food safety with disease surveillance data.

D. Broader Impact and Societal Significance

This research contributes to UN Sustainable Development Goals by enabling early contamination detection and reducing foodborne illnesses (SDG 3) [4], minimizing food waste through accurate quality monitoring (SDG 12) [8], and facilitating collaboration between regulators, manufacturers, and consumers (SDG 17) [9]. Beyond food safety, the methodological contributions around temporal knowledge graphs, domain-adaptive language models, and multi-modal data fusion apply to healthcare monitoring, environmental risk assessment, and supply chain security [6], [15], and [16].

E. Concluding Remarks

Traditional manual food safety monitoring cannot keep pace with modern supply chain complexity [9]. This research demonstrates that AI-driven approaches integrating computer vision, NLP, temporal knowledge graphs, and deep learning can transform food safety from reactive inspection to proactive prediction [1], [2], [6], [7]. The system achieves 85% reduction in manual analysis time, 86.15% accuracy exceeding static methods by 21.92%, sub-5-second event detection latency, and interpretable risk assessments supporting regulatory decisions. While challenges remain around multilingual support and emerging contaminant detection [15], [17], the architecture provides a scalable foundation for continuous improvement. As food supply chains grow increasingly global, intelligent monitoring systems become essential infrastructure for public health protection [4], [13].

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